

Language teachers' AI literacy: A psychometric study based on the ED-AI framework

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ABSTRACT

Artificial Intelligence (AI) is reshaping language education, making AI literacy crucial for teachers to engage critically and effectively with this technology. Nonetheless, most existing assessments target students or general users, leaving a gap in measuring AI literacy within language teacher education. This study sought to develop and validate the Teachers' AI Literacy Scale (TAILS), grounded in the ED-AI literacy framework, which comprises six dimensions: knowledge, evaluation, collaboration, contextualization, autonomy, and ethics. The scale was tested with preservice English language teachers through two phases: exploratory factor analysis (EFA) with 165 participants and confirmatory factor analysis (CFA) with a separate sample of 227. Results confirmed a six-factor structure with high internal consistency (Cronbach's α values > 0.90) and acceptable model fit indices (Chi-square/df = 1.766, RMSEA = 0.058, SRMR = 0.054, TLI = 0.908, CFI = 0.919), demonstrating strong validity and reliability. Each dimension aligned clearly with the competencies required for AI-integrated language teaching. The TAILS is a psychometrically robust, context-specific instrument for assessing AI literacy in language teacher education. This study bridges the gap between theoretical frameworks and practical assessment, offering a foundation for curriculum development, professional training, and policymaking. Its application supports the preparation of AI-competent educators equipped to navigate the ethical, pedagogical, and technological demands of the digital classroom.

1. Introduction

Artificial Intelligence (AI) has become a transformative force across diverse fields, encompassing education. Its rapid integration into language learning and teaching highlights the urgent need for a deeper understanding of AI literacy, particularly in language teacher education. Language teachers are now expected to integrate AI tools effectively, foster positive attitudes, deliver content efficiently, and maintain overall instructional quality (Chu et al., 2023). AI literacy refers to the ability to understand, utilize, evaluate, and critically reflect on AI technologies and their implications, enabling meaningful interaction with AI systems (Chiu et al., 2024). For language teachers, this literacy is crucial for navigating AI-driven instructional tools, facilitating personalized learning experiences, providing real-time feedback on language proficiency, and offering authentic opportunities for language

practice—ultimately enhancing student engagement and language acquisition (Fu et al., 2020; Kim et al., 2021). Generative AI further empowers educators to create customized lesson plans and generate assessment tasks aligned with learning objectives more efficiently (Shah, 2023). However, despite the growing centrality of AI literacy in educational discourse, its conceptualization and assessment remain insufficiently theorized for the specific epistemic, pedagogical, and ethical demands of language teacher education.

Importantly, AI literacy in language teacher education cannot be treated as equivalent to general AI literacy. While general AI literacy frameworks typically emphasize technological awareness, functional use, and attitudinal dispositions toward AI (Ng et al., 2021; Wang et al., 2022), language teachers operate within a discipline that is inherently communicative, interpretive, and socially situated (Mont et al., 2026; Torres & Kahveci, 2025). Language teaching requires teachers to engage

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in continuous pedagogical judgment related to meaning-making, interaction, feedback, cultural sensitivity, and learner identity, processes that are increasingly shaped by AI systems (AlAfnan, 2024; Chen, 2026). As facilitators of communicative practice rather than mere users of technology, language teachers must evaluate AI-generated language output, mediate its pedagogical appropriateness for specific learners and contexts, and balance automation with human interaction (Chiu et al., 2024; Chu et al., 2023). These responsibilities call for a discipline-specific AI literacy profile that foregrounds pedagogical agency, contextual adaptation, collaborative orchestration, and ethical decision-making. Such dimensions extend beyond the scope of generic AI literacy constructs and underscore the need for a specialized assessment framework in language teacher education.

Although existing research underscores the significance of AI literacy in enhancing pedagogical practices and enriching student learning experiences (Boscardin et al., 2023; Ibrahim, 2024), multiple AI literacy frameworks and instruments have primarily been developed for students, nonexperts, or general users. For instance, the Artificial Intelligence Literacy Questionnaire (AILQ) assesses affective, behavioral, cognitive, and ethical dimensions (Ng et al., 2023), while the AI Literacy Scale (AILS) focuses on awareness, usage, evaluation, and ethics (Wang et al., 2022). Other instruments, such as the Meta AILS (MAILS), AI Anxiety Scale (AIAS), and nonexpert-focused frameworks (Carolus et al., 2023; Laupichler et al., 2023; Wang & Wang, 2022), similarly privilege general technological engagement or psychological responses to AI. Conceptually, these instruments fall short in capturing teachers' pedagogical agency, contextual decision-making, and professional autonomy; practically, they risk reducing AI literacy to generic competencies that fail to inform instructional design, ethical classroom use, or teacher professional development. As a result, the absence of a discipline-sensitive assessment framework obscures how language teachers interpret, adapt, and govern AI use within instructional contexts.

This shortcoming constitutes a clear theoretical gap (Lim, 2026). While AI literacy is widely acknowledged as essential, existing scholarship inadequately theorizes how AI literacy manifests as a professional competence in language teacher education, and how it differs from learner- or user-oriented conceptions. When this gap is left unaddressed, AI literacy remains under-specified as a construct, limiting both theoretical advancement and practical applicability. Teachers may be evaluated using instruments that are misaligned with their instructional realities, leading to weak construct validity, superficial professional insights, and poorly targeted teacher education interventions.

While prior studies have emphasized the role of AI in optimizing language learning (Gyawali & Mehandroo, 2022), limited empirical work has systematically examined AI literacy as a multidimensional, pedagogically grounded construct among language teachers themselves. This lack of theory-informed measurement hinders the field's ability to assess teachers' preparedness, compare competency profiles, or examine how AI literacy relates to instructional decision-making and ethical practice. In theoretical terms, the problem lies not merely in the absence of an instrument, but in the absence of a construct specification that reflects teachers' situated roles as designers, evaluators, collaborators, and ethical agents in AI-mediated language education.

To address this gap, the present study develops and validates the Teachers' AI Literacy Scale (TAILS) for language teacher education. Grounded in the ED-AI literacy framework (Allen & Kendeou, 2024), TAILS conceptualizes AI literacy across six theoretically distinct yet interrelated dimensions: knowledge, evaluation, collaboration, contextualization, autonomy, and ethics. Theoretically, this study contributes by adapting and extending AI literacy theory to a discipline-specific professional context, thereby advancing construct clarity and contextualized theorization (Lim, 2026). This represents an incremental yet meaningful theoretical extension, as it shifts AI literacy from a generalized competence model toward a profession- and pedagogy-sensitive framework that foregrounds teacher agency and ethical responsibility.

The theoretical interestingness of this study lies in its reframing of AI literacy as a pedagogical and ethical capacity rather than a purely technical one, contextualized in language teacher education. By systematically operationalizing this reconceptualization through psychometric validation, the study bridges theory development and empirical testing. Practically, the validated TAILS instrument enables language educators, teacher education programs, and policymakers to diagnose AI literacy competencies, design targeted professional development, and guide responsible AI integration in language education.

2. Literature review

2.1. Defining AI literacy

AI literacy refers to the ability to understand, utilize, evaluate, and critically analyze AI technologies and their impacts, allowing individuals to effectively interact with AI systems and apply them across multiple contexts (Chiu et al., 2024). Core components of AI literacy include the capacity to critically assess AI technologies, communicate with AI systems, and collaborate effectively in environments where AI plays a significant role (Fan & Zhang, 2024; Ng et al., 2021). Despite general agreement on its importance, definitions and frameworks typically overlap, highlighting the need for a diverse yet cohesive educational approach to AI (Otero et al., 2023). Additionally, some studies advocate for integrating AI literacy into existing curricula at all educational levels—from early childhood and K–12 to higher education—to foster a more comprehensive development of AI competencies (Otero et al., 2023; Relmasira et al., 2023; Yetişensoy & Rapoport, 2023).

The call for educators to optimize their AI literacy is increasingly urgent. As stewards of AI literacy, educators must be equipped to effectively guide students in navigating the complexities of AI (Boscardin et al., 2023; Ibrahim, 2024). Numerous studies emphasize the significance of teacher preparedness, recommending targeted training and curriculum development to ensure educators possess the necessary skills to effectively integrate AI tools into their teaching practices (Yeter et al., 2024; Zhao et al., 2022). Programs aimed at developing AI literacy must strike a balance between technical proficiency and ethical awareness. Students need to learn not only how to utilize AI tools but also how to critically evaluate their broader societal implications (Biagini, 2024; Choi et al., 2024). This balanced approach seeks to cultivate responsible AI users who are aware of the ethical dimensions and long-term consequences of their engagement with AI technologies.

In the context of English language education, AI literacy involves the knowledge and ability to effectively engage with AI technologies that influence language learning and teaching practices. Research highlights that AI can significantly refine the quality of English language acquisition by promoting personalized learning experiences (Gyawali & Mehandroo, 2022). Developing AI literacy goes beyond understanding how AI tools function; it also requires the cultivation of critical digital skills to navigate these technologies effectively. Recent studies underscore the importance of critical digital literacy—particularly in relation to AI—as it encourages learners to engage with AI tools analytically, strengthening their cognitive development and critical thinking skills (Joseph, 2023; Wang et al., 2024). This competency is particularly crucial as learners increasingly interact with advanced applications, such as ChatGPT, which can significantly shape their learning processes (Joseph, 2023; Wei, 2023).

Furthermore, AI literacy fosters a deeper understanding of the ethical considerations surrounding the use of AI in education. As AI technologies become more integrated into language learning environments, concerns have emerged regarding inclusivity, fairness, and the ethical implications of relying on automated systems (AlAfnan, 2024). Thus, knowledge about AI must transcend technical skills and include an awareness of its cultural and societal impacts (AlAfnan, 2024). This broader perspective allows both educators and learners to critically

examine the appropriateness and effectiveness of AI tools within their instructional contexts, encouraging a more holistic and responsible approach to language education.

2.2. AI literacy assessment tools

AI literacy tools comprise a diverse array of resources, frameworks, and educational approaches designed to equip individuals with the knowledge and skills required to understand, utilize, and critically engage with AI technologies. As AI becomes increasingly integrated across sectors, encompassing education, the development of AI literacy tools has garnered significant attention from researchers and educators.

To date, several scales have been developed to assess AI literacy. A notable example is the Artificial Intelligence Literacy Questionnaire (AILQ), designed to assess AI literacy among secondary students (Ng et al., 2023). The AILQ evaluates four key dimensions: affective learning, which encompasses intrinsic motivation and self-efficacy; behavioral learning, which comprises commitment and collaboration; cognitive learning, which covers knowledge acquisition and application; and ethical considerations. By incorporating these domains, the AILQ offers a holistic view of AI literacy that integrates both technical competence and ethical awareness. Similarly, Laupichler et al. (2023) developed an AI literacy assessment for nonexperts, emphasizing technical knowledge, critical evaluation, and practical application—making it especially relevant for individuals without formal AI training. This assessment prioritizes technical knowledge, critical evaluation, and practical application, making it particularly relevant for individuals without formal AI training. Additionally, an Artificial Intelligence Literacy Scale (AILS) for primary and middle school teachers was also developed according to a set of criteria established by Ng et al. (2021). This scale includes four dimensions: knowing and understanding AI, applying AI, evaluating AI application, and AI ethics (Zhao et al., 2022).

Beyond literacy assessment, other tools have been developed to explore the psychological and emotional dimensions of AI engagement. The AI Anxiety Scale (AIAS) measures individuals' anxiety and attitudes toward AI technologies, acknowledging the psychological impact of AI interactions (Wang & Wang, 2022). Another notable instrument is the Meta AILS (MAILS), recognized for its modular design, which allows for tailored assessments across diverse audiences (Carolus et al., 2023). This tool integrates psychological components, such as problem-solving and emotion regulation, alongside AI-related competencies. Likewise, the AI Literacy Scale (AILS) assesses AI literacy among general users, focusing on awareness, usage, evaluation, and ethical considerations (Wang et al., 2022). Additionally, several tools have been designed to assess public attitudes toward AI. The Attitude Toward Artificial Intelligence (ATAI) scale evaluates emotional and attitudinal barriers to AI engagement and has been validated across populations in Germany, China, and the UK (Sindermann et al., 2021). Similarly, the General Attitudes Toward Artificial Intelligence Scale (GAAIS) examines public perceptions of AI by distinguishing between acceptance and fear through its positive and negative subscales (Schepman & Rodway, 2020). Collectively, these tools underscore the significance of evaluating not only technical expertise but also ethical awareness and socio-emotional capacities. This comprehensive perspective reflects the evolving landscape of AI literacy and highlights the need for educational approaches that address its multifaceted nature.

Although these instruments have contributed substantially to the field, most were designed for students, general users, or nonexpert populations and therefore prioritize general technological awareness, affective responses, or broad AI competencies. As a result, they offer limited insight into teachers' pedagogical agency, contextual decision-making, and professional responsibility when integrating AI into subject-specific instruction. In particular, they do not sufficiently capture how educators evaluate AI-generated content, adapt AI tools to instructional goals and learner needs, collaborate with AI in teaching practices, or navigate ethical challenges within classroom contexts. To

clarify these differences and to situate the present study, Table 1 provides a comparative overview of key AI literacy measurement instruments and contrasts them with the proposed TAILS. This comparison highlights differences in intended audience, dimensions assessed, and key limitations, thereby underscoring the need for a discipline-specific, pedagogy-oriented instrument for language teacher education.

2.3. ED-AI literacy framework and this study

This research seeks to achieve two main goals: (1) to design a self-assessment tool for measuring language teachers' AI literacy and (2) to examine the instrument's reliability and validity. It is grounded in the novel ED-AI literacy framework, which includes six dimensions: knowledge, evaluation, collaboration, contextualization, autonomy, and ethics (Allen & Kendeou, 2024). The conceptual definitions of these dimensions are presented in Table 2.

The selection of this framework was guided by the "IMPACT" criteria for theory selection, including *Interestingness*, *Matching*, *Parsimony*, *Applicability*, *Conceptual rigor*, and *Testability* (Hollebeek et al., 2025). In terms of *Interestingness*, the ED-AI framework advances AI literacy research by extending beyond technical knowledge (knowledge) to foreground teachers' critical judgment (evaluation), professional agency

Table 1

Comparison of existing AI literacy measurement instruments and TAILS.

Instrument	Intended audience	Core dimensions assessed	Key limitations for teacher education
AILG (Ng et al., 2023)	Secondary students	Affective, behavioural, cognitive, ethics	Learner-focused; limited pedagogical and instructional relevance
AILS (Wang et al., 2022)	General users	Awareness, usage, evaluation, ethics	Generic AI use, not profession or discipline specific
AILS-Teachers (Zhao et al., 2022)	Primary & middle school teachers	Knowing AI, applying AI, evaluating AI, ethics	Broad teachers' focus; limited contextual and subject specificity
AI literacy of non-experts (Laupichler et al., 2023)	Nonexperts	Technical knowledge, evaluation, application	Emphasizes technical competence over pedagogy
MAILS (Carolus et al., 2023)	General users	Apply AI, understand AI, detect AI, AI ethic, create AI, AI self-efficacy (AI problem solving, AI learning), AI self-competency (AI persuasion literacy, AI emotion regulation)	Focuses on meta-cognition and emotions rather than teaching practices
AIAS (Wang & Wang, 2022)	General users	Learning, AI configuration, job replacement, sociotechnical blindness.	Measures AI anxiety or affective responses only
GAAIS (Schepman & Rodway, 2020).	General public	Positive and negative general attitudes toward AI	Captures general AI attitudes rather than teaching-specific perspectives.
ATAI scale (Sindermann et al., 2021).	Population of Germany, China, and the UK	Attitudes towards AI of acceptance and fear dimensions.	Captures AI attitudes rather than pedagogical and instructional relevance
TAILS (current study)	Pre-service language teachers	Knowledge, evaluation, collaboration, contextualization, autonomy, ethics	Designs to capture pedagogical, contextual, and ethical dimensions of AI use in language teacher education

Table 2
Conceptual definition of the TAILS dimensions (Adapted from Allen & Kendeou, 2024).

Dimensions of TAILS	Conceptual definition
Knowledge	Teachers' understanding of fundamental AI concepts, tools, and their applications within educational contexts
Evaluation	Teachers' ability to critically analyze and compare various AI tools for educational purposes.
Collaboration	The skill to effectively integrate AI tools to improve instructional design and foster student engagement
Contextualization	The capacity to adapt and tailor AI applications to specific subject areas, diverse student populations, and cultural settings
Autonomy	The competence to independently manage and operate AI technologies in educational settings.
Ethics	The awareness and application of ethical principles related to AI use in education.

(autonomy), and ethical responsibility (ethics), offering a more pedagogically meaningful and discipline-sensitive conceptualization for language education. Regarding *Matching*, the framework aligns closely with the construct of language teachers' AI literacy because its dimensions reflect core teaching practices, including informed tool use (knowledge), pedagogical decision-making (evaluation), instructional integration (collaboration), and adaptation to subject-specific, cultural, and learner-related conditions (contextualization). With respect to *Parsimony*, the six dimensions collectively capture the essential competencies required for educational AI use without conceptual redundancy, balancing cognitive, practical, and ethical aspects of teachers' AI engagement. In terms of *Applicability*, each dimension maps directly onto teacher education and classroom practice, supporting curriculum design, instructional planning, professional development, and responsible AI implementation in diverse educational contexts. Regarding *Conceptual rigor*, the framework provides clearly defined, theoretically grounded, and internally coherent dimensions, enabling precise construct specification and systematic scale development. Finally, in terms of *Testability*, the dimensions are sufficiently concrete and operationalizable, ranging from teachers' understanding (knowledge) and critical appraisal (evaluation) to independent use (autonomy) and ethical awareness (ethics) allowing them to be empirically examined through established psychometric procedures.

3. Methods

3.1. Research design

This study employed a cross-sectional design to construct and validate the instrument, carried out in two primary phases: item development and psychometric assessment, to ensure both validity and reliability (Boateng et al., 2018; DeVellis & Thorpe, 2021; Lim, 2024; Yusoff et al., 2021). Specifically, the research was conducted in two phases: (1) pre-data-collection validation and (2) post-data-collection psychometric evaluation. In Phase 1, an initial pool of items was generated through an extensive review of relevant literature and theoretical frameworks. Content validity and face validity were established as prerequisite forms of validity, with content validity assessed through expert review and face validity evaluated via a pilot study with the target population to ensure item clarity, relevance, and comprehensibility. Feedback from this phase informed iterative refinement of the instrument. Phase 2 focused on psychometric evaluation, including exploratory and confirmatory factor analyses to examine construct validity (convergent and discriminant validity) and internal consistency reliability.

3.1.1. Phase I: Instrument development

3.1.1.1. Step 1: Identifying the framework and generating scale items. The TAILS was developed according to the ED-AI literacy framework, which comprises six dimensions: knowledge, evaluation, collaboration, contextualization, autonomy, and ethics (Allen & Kendeou, 2024). The conceptual foundation of the scale was further shaped by a synthesis of empirical and theoretical contributions to the field of AI literacy (Ayanwale et al., 2024; Liu et al., 2023; Long & Magerko, 2020; Ng et al., 2021; Zhao et al., 2022).

To ensure systematic alignment between the theoretical framework and the generated items, we employed a formal item-to-construct mapping procedure during the item development stage. Specifically, an initial item pool was created for each ED-AI dimension based on its conceptual definition, and each item was explicitly mapped to a single target construct. This mapping matrix documented the theoretical rationale for item inclusion and ensured that all dimensions were adequately represented without conceptual overlap.

In this research, scale items were generated based on a comprehensive literature review to ensure alignment with the framework and the instructional realities of language teacher education. Following initial item generation, an expert agreement matrix was used to evaluate the correspondence between items and their intended constructs. Subject-matter experts in teacher education and educational technology independently reviewed each item and indicated its relevance to the corresponding ED-AI dimension. Items were retained, revised, or discarded based on consensus regarding construct alignment and clarity. This process strengthened content validity and reduced the risk of construct contamination.

The final scale comprises 30 items distributed across six factors, each capturing a distinct dimension of AI literacy. The knowledge factor (five items) assesses participants' understanding of AI in education. For instance, "I know that AI can help teachers create personalized learning for students." The evaluation factor (five items) measures the ability to assess AI tools. For instance, "I am capable of comparing the effectiveness of different AI tools for classroom use." The collaboration factor (five items) examines the ability to work effectively alongside AI in teaching. For instance, "I can collaborate with AI tools to design engaging lessons." The contextualization factor (five items) evaluates the capacity to integrate AI into specific instructional contexts. For instance, "I can integrate AI tools into lesson plans to enhance specific subjects, such as language learning." The autonomy factor (five items) prioritizes decision-making concerning AI use. For instance, "I can decide when it is appropriate to use AI tools in my teaching and when not to." Lastly, the ethics factor (five items) assesses ethical considerations in using AI. For instance, "I actively consider students' privacy when using AI tools." Participants responded using a five-point Likert scale (1 = strongly disagree, 2 = disagree, 3 = neither agree nor disagree, 4 = agree, and 5 = strongly agree). Additionally, the questionnaire encompassed demographic items, such as gender, academic level, and teaching experience, to offer further contextual insights into the participants' backgrounds.

3.1.1.2. Step 2: Testing for content validity. Content validity refers to the degree to which an assessment instrument effectively captures the full scope and depth of the intended construct (Taherdoost, 2016). To ensure content validity, a mixed quantitative–qualitative expert review process was employed. In addition to conducting a comprehensive literature review, we engaged seven expert reviewers from the educational field to examine the instrument. Specializing in teacher education, these experts assessed and rated each quality indicator through an online platform. Quantitatively, the Content Validity Index (CVI) was used to operationalize expert judgments. At the item level, the item-level CVI (I-CVI) was determined by calculating the proportion of experts who rated a quality indicator's relevance as three or higher on a 4-point Likert scale

(1 = totally irrelevant, 2 = irrelevant, 3 = relevant, and 4 = totally relevant). At the dimension level, the scale-level CVI (S-CVI) was computed as the average of the I-CVI scores across all items within a given dimension. According to Polit et al. (2007), a scale is considered to have excellent content validity when each item attains an I-CVI of at least 0.78 and the scale-level content validity index/averaging method (S-CVI/Ave) reaches 0.90 or higher. The scale-level content validity index and universal agreement (S-CVI/UA) were calculated. Additionally, the modified kappa (κ^*) was employed to adjust for potential chance agreement among experts, with values of at least 0.75 indicating acceptable levels of agreement (Polit et al., 2007).

Qualitatively, experts provided written feedback addressing terminological appropriateness, linguistic clarity, conceptual alignment with the ED-AI literacy framework, and potential ambiguities in item phrasing. This qualitative feedback served a complementary function to the CVI statistics by identifying areas where items could be improved for clarity and interpretability without altering their conceptual intent. The integration of quantitative CVI indices and qualitative expert commentary ensured that the instrument was both conceptually representative and clearly articulated, thereby strengthening its substantive foundation prior to pilot testing and subsequent psychometric validation.

3.1.1.3. Step 3: Pilot study for face validity. To further evaluate the suitability and usability of the scale, a pilot study was conducted to establish face validity as an early-stage form of validity evidence focused on participants' intuitive understanding of the instrument. Face validity refers to the extent to which a measure appears, on the surface, to assess the intended construct, as judged by nonexperts such as test-takers. It focuses on aspects such as feasibility, readability, style consistency, and clarity of language (Taherdoost, 2016). The pilot study involved 30 preservice English language teachers in Indonesia, a sample size that is appropriate for face validity assessment and justified by the principle of data saturation, whereby additional participants no longer yield substantively new feedback regarding item clarity, comprehensibility, or relevance (Lim, 2024). During the pilot phase, no substantively new concerns emerged after successive rounds of participant feedback, indicating that saturation had been reached. In addition, response time and ease of completion were assessed to ensure operational feasibility. Although participants reported no major difficulties in understanding or responding to the items, they provided minor suggestions aimed at improving wording precision and stylistic consistency, which were incorporated into the revised instrument.

3.1.2. Phase II: Validation of the scale (psychometric property testing)

3.1.2.1. Step 4: Construct validity. Construct validity was rigorously assessed through a two-stage factor-analytic approach, comprising exploratory factor analysis (EFA) followed by confirmatory factor analysis (CFA). In the initial step, EFA was conducted using principal component analysis (PCA) with varimax rotation. PCA was selected at this stage as a data-reduction technique appropriate for early-scale development, where the primary objective was to explore the dimensional structure and identify a parsimonious factor solution (Yusoff et al., 2021). To ensure the data's suitability for EFA, the Kaiser–Meyer–Olkin (KMO) measure and Bartlett's test of sphericity were conducted, verifying the adequacy of the sample and the correlation between variables. The number of factors retained was determined on the basis of multiple criteria, encompassing eigenvalues, scree plot inspection, factor loadings, and the percentage of explained variance (Yusoff et al., 2021). These criteria were crucial for identifying a meaningful and robust factor structure, ensuring that the instrument's dimensions were properly represented. Additionally, a robustness check was conducted. Specifically, the EFA was re-estimated using a principal axis factoring with promax rotation, which allows factors to correlate and is theoretically consistent with the multidimensional nature of AI

literacy.

Following EFA, the factor structure was further validated using CFA. CFA was estimated using covariance-based structural equation modeling within SmartPLS 4. Although SmartPLS is widely recognized for partial least squares estimation, the current analysis employed covariance-based model estimation procedures supported by SmartPLS 4, using maximum likelihood estimation. Model fit evaluation emphasized Comparative Fit Index (CFI) and Tucker–Lewis Index (TLI) values near or above 0.90, alongside Root Mean Square Error of Approximation (RMSEA) values below 0.08. The chi-square to degrees of freedom ratio (χ^2/df) was also examined, with values below 5 indicating acceptable fit (Hu & Bentler, 1999; Yusoff et al., 2021). Goodness of Fit Index (GFI) and Adjusted Goodness of Fit Index (AGFI) were treated as supplementary diagnostics rather than primary decision rules.

3.1.2.2. Step 5: Internal consistency testing. The internal consistency reliability of the instrument was assessed using Cronbach's alpha, with values calculated separately for each factor to ensure that each dimension demonstrated sufficient reliability. Cronbach's alpha is a widely employed statistic for evaluating the degree of consistency within a set of items measuring the same construct. Calculating it separately for each factor enabled us to assess the reliability of each specific dimension. According to Taherdoost (2016), the recommended thresholds for newly developed instruments are as follows: excellent (0.90 and above), high (0.70–0.90), moderate (0.50–0.70), and low (0.50 and below). These benchmarks provide a standard for determining whether the reliability of the instrument falls within an acceptable range. In addition to Cronbach's alpha, we also calculated the composite reliability for each dimension, which offers a more comprehensive measure of internal consistency by considering the relationships among all items within a particular factor.

3.2. Context and participants

This research involved preservice English language teachers enrolled in universities offering English language teacher education programs in Indonesia. Participants were selected through convenience sampling, a method chosen for its practicality, cost-effectiveness, and efficiency compared with other sampling techniques. Additionally, eligibility criteria encompassed participants' willingness to take part in the study and prior experience teaching English. This population was considered particularly appropriate for the intended use of the TAILS because preservice teachers constitute a critical transitional group, simultaneously developing pedagogical knowledge, professional identity, and dispositions toward the pedagogically responsible integration of AI technologies. As such, the study's research design, sampling strategy, and validation claims are intentionally bounded to this population and educational context. Consequently, interpretations of the findings and applications of TAILS beyond comparable preservice English language teacher education settings should be made with caution and supported by further validation in alternative populations and contexts.

A total of 422 preservice teachers participated in the research, distributed across distinct stages of instrument validation. 30 individuals were involved in the pilot study to refine and assess the clarity of the instrument, 165 participated in EFA to identify the underlying factor structure, and 227 were assessed for CFA to validate the proposed model and confirm the instrument's reliability and validity. Specifically, Table 3 exhibits the descriptive statistics of the participants: sample 1 (N = 165) for EFA and sample 2 (N = 227) for CFA. In sample 1, most participants were female (77.6%), while males comprised 22.4%. Most participants were aged 18–20 years (62.4%), followed by those aged 21–23 years (33.3%), a small group aged 24–26 years (3.6%), and only one participant aged 27 or older (0.6%). Regarding academic level, most were in their second year (37.0%), followed by the third year (35.8%), fourth year or higher (19.4%), and first year (7.9%). The majority

Table 3
Descriptive statistics.

Variable	Categories	Sample 1 (N = 165)	Sample 2 (N = 227)
		N (%)	N (%)
Gender	Male	37 (22.4%)	48 (21.1%)
	Female	128 (77.6%)	179 (78.9%)
Age	18–20 years	103 (62.4%)	124 (54.6%)
	21–23 years	55 (33.3%)	97 (42.7%)
	24–26 years	6 (3.6%)	6 (2.6%)
	27 years and more	1 (0.6%)	-
Academic level	First year	13 (7.9%)	49 (21.6%)
	Second year	61 (37.0%)	52 (22.9%)
	Third year	59 (35.8%)	85 (37.4%)
	Fourth year and more	32 (19.4%)	41 (18.1%)
Teaching experiences	1–2 years	161 (97.6%)	223 (98.2%)
	3–4 years	2 (1.2%)	2 (0.9%)
	7 years and more	2 (1.2%)	2 (0.9%)

(97.6%) reported having 1–2 years of teaching experience, with only a few indicating 3–4 years or 7 years or more. In Sample 2, the demographic profile was similarly female-dominated (78.9%), with males accounting for 21.1%. The age distribution was comparable, with most participants aged 18–20 years (54.6%), followed by those aged 21–23 years (42.7%), and a small portion aged 24–26 years (2.6%); no participants were aged 27 years or above. Concerning academic year, the largest group was in their third year (37.4%), followed by the second year (22.9%), the first year (21.6%), and the fourth year or higher (18.1%). Almost all participants (98.2%) had 1–2 years of teaching experience, with very few reporting longer durations. Overall, both samples comprised predominantly young, female preservice teachers with minimal teaching experience, reflecting a consistent demographic profile across the two groups.

3.3. Data collection and ethical considerations

Prior to the initiation of the study, a formal request for ethical approval was submitted to the Institutional Review Board of the university. The research received approval under reference number 1/2025, ensuring adherence to established ethical standards for research involving human participants. Data were collected in the spring of 2024 via an online survey, providing a convenient and efficient method for participant engagement. All responses were treated as strictly confidential, and strict measures were taken to protect participants' privacy. Participation in the study was entirely voluntary, with respondents providing informed consent before participation. They were clearly informed that they had the right to withdraw from the study, ensuring that their participation was completely voluntary and based on their informed decision.

3.4. Data analysis

Data analysis was conducted using SPSS 26 for descriptive statistics and EFA, and SmartPLS 4 for CFA and measurement model evaluation. SmartPLS 4 was employed as the analytical platform due to its flexible estimation capabilities, including covariance-based structural equation modeling. Descriptive statistics were initially employed to summarize the quantitative data, including sample size, percentages for categorical variables, means, and standard deviations. This step provided a foundational understanding of the data distribution and participant characteristics. To evaluate the structural validity of the questionnaire, both EFA and CFA were applied. EFA was employed to explore the underlying factor structure of the instrument and determine the appropriate number of factors, while CFA was conducted to validate the factor structure identified through EFA and assess the goodness-of-fit of the model.

4. Results

4.1. Content validity

Content validity was established through both quantitative indices and qualitative expert feedback. Seven experts evaluated the content of each quality indicator using an online platform. Quantitatively, the I-CVI scores for all 30 items ranged from 0.86 to 1.00, indicating strong content validity. The S-CVI for the six dimensions ranged from 0.94 to 1.00, demonstrating a high level of agreement among experts within each dimension. The S-CVI/Ave for the entire scale was 0.98, exceeding the threshold for excellent content validity. The S-CVI/UA was 0.87, specifying that most items were deemed relevant by most experts. According to Polit et al. (2007), content validity is considered excellent when each item achieves an I-CVI of 0.78 or higher, and the scale attains an S-CVI/Ave of 0.90 or above. The I-CVI evidence was supplemented with the modified kappa (κ^*) statistic to account for chance agreement among experts. All items demonstrated strong to perfect chance-corrected agreement, with modified kappa values ranging from 0.86 to 1.00, exceeding the recommended threshold of at least 0.75 (Polit et al., 2007), thereby indicating robust expert consensus beyond chance. Appendix 2 reports item-level content validity index and modified kappa values for all items, supporting the content validity conclusion.

Additionally, expert reviewers provided detailed written feedback on item wording, terminological precision, and alignment with the ED-AI literacy framework dimensions. The most frequent suggestions concerned improving linguistic clarity, reducing potential ambiguity, and ensuring consistency in the use of AI-related terminology across items. Several experts also recommended minor rephrasing to better reflect pedagogical practices specific to language teacher education and to avoid overly technical expressions that could hinder comprehension by preservice teachers. Based on this qualitative feedback, six items were reworded to enhance clarity, readability, and conceptual alignment, while no items were deleted or added at this stage.

4.2. Exploratory factor analysis

The initial assessment of construct validity was conducted through EFA using a sample of 165 participants. The Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy was 0.906, revealing that the dataset was highly suitable for factor analysis. Additionally, Bartlett's test of sphericity produced an approximate chi-square value of 2737.390 with 435 degrees of freedom, which was highly significant ($p < 0.001$), validating that the item correlations were strong enough to proceed with factor analysis. Using the Kaiser criterion, analysis of the scree plot (see Fig. 1), and evaluation of the proportion of explained variance, the study identified six distinct factors aligned with the framework proposed by Allen and Kendeou (2024).

In the initial EFA, PCA with varimax rotation was employed as an exploratory data-reduction approach to identify the underlying dimensional structure of the scale. To address concerns regarding the theoretically expected correlations among dimensions of AI literacy, a robustness analysis was subsequently conducted. Specifically, the EFA was re-estimated using principal axis factoring with promax rotation, which allows latent factors to correlate and is conceptually aligned with the multidimensional and interrelated nature of AI literacy constructs. The robustness analysis yielded the same six-factor structure, with comparable item–factor loadings and consistent item composition across factors, indicating that the identified structure was stable and not dependent on the initial extraction or rotation method.

As exhibited in Table 4, the factor analysis revealed six distinct factors aligned with the original AI literacy framework, each comprising items with satisfactory factor loadings. The first factor—*Knowledge*—encompassed five items with loadings ranging from 0.417 to 0.802, an eigenvalue of 1.793 accounting for 5.976% of the variance,

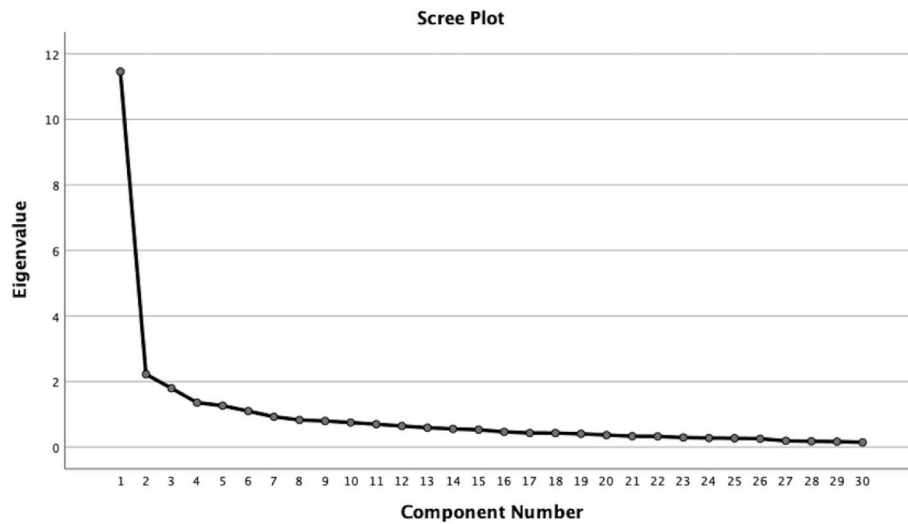


Fig. 1. Scree plot (N = 165).

Table 4

Exploratory factor analysis of the TAILS (N = 165).

Codes	Factors and items	Factors					
		1	2	3	4	5	6
Knowledge (KNO)							
KNO1	1. I know that AI can help teachers create personalized learning for students			0.738			
KNO2	2. I am familiar with some common AI tools that support teaching and learning in classrooms (e.g., ChatGPT).			0.417			
KNO3	3. I know that AI can support students with different learning needs			0.562			
KNO4	4. I know that AI can assist teachers in lesson planning and material development			0.801			
KNO5	5. I am aware that AI applications can help teachers with assessment			0.802			
Evaluation (EVA)							
EVA1	6. I am capable of comparing the effectiveness of different AI tools for classroom use.					0.718	
EVA2	7. I can select the most appropriate solution from different options provided by AI applications.					0.478	
EVA3	8. I can choose the right educational AI tools based on the specific task.					0.666	
EVA4	9. I can evaluate how AI recommendations align with curriculum standards and goals.					0.804	
EVA5	10. I can analyze whether AI assessment tools for students are accurate and fair.					0.648	
Collaboration (COL)							
COL1	11. I can collaborate with AI tools to design engaging lessons.						0.579
COL2	12. I can use AI to support teamwork among students in group projects.						0.599
COL3	13. I can communicate effectively with AI tools to modify group learning activities based on student needs.						0.528
COL4	14. I feel comfortable integrating feedback from AI systems into collaborative lesson planning.						0.600
COL5	15. I can facilitate student collaboration using AI-enhanced platforms (e.g., Padlet).						0.442
Contextualization (CON)							
CON1	16. I can integrate AI tools into lesson plans to enhance specific subjects, such as language learning.				0.399		
CON2	17. I understand how to use AI to support students with specific needs				0.461		
CON3	18. I can design classroom activities that encourage students to explore AI technologies.				0.546		
CON4	19. I can apply AI tools to improve teaching strategies.				0.567		
CON5	20. I can adapt AI applications to suit cultural contexts in my teaching.				0.631		
Autonomy (AUT)							
AUT1	21. I can decide when it is appropriate to use AI tools in my teaching and when not to.	0.592					
AUT2	22. I am capable of learning new AI tools independently for teaching purposes.	0.683					
AUT3	23. I can troubleshoot basic technical issues with AI tools used in the classroom.	0.794					
AUT4	24. I am confident in guiding students on how to interact with AI responsibly.	0.644					
AUT5	25. I feel prepared to design lessons without relying solely on AI, when necessary.	0.457					
Ethics (ETH)							
ETH1	26. I actively consider students' privacy when using AI tools.		0.794				
ETH2	27. I prioritize ethical considerations when selecting AI tools for classroom use.		0.754				
ETH3	28. I can quickly identify ethical issues when using AI applications for my teaching.		0.596				
ETH4	29. I can identify and address potential fairness issues when using AI for assessment.		0.715				
ETH5	30. I feel responsible for ensuring that AI use in my teaching activities is ethical.		0.704				
Eigenvalue		11.458	2.224	1.793	1.355	1.262	1.098
Percentage of variance explained (%)		38.194	7.412	5.976	4.517	4.207	3.659
Cumulative percentage of variance explained by the factor model (%)		38.194	45.605	51.582	56.099	60.305	63.965
Cronbach's alpha		0.829	0.867	0.820	0.851	0.791	0.759

and a Cronbach's alpha of 0.820. The second factor—*Evaluation*—comprised five items with loadings between 0.478

and 0.804, an eigenvalue of 1.262 explaining 4.207% of the variance, and a Cronbach's alpha of 0.791. The third factor—*Collaboration*—contained five items with loadings from 0.442 to 0.600, an eigenvalue of 1.098 contributing 3.659% of the variance, and a Cronbach's alpha of 0.759. The fourth factor—*Contextualization*—consisted of five items with loadings ranging from 0.399 to 0.631, an eigenvalue of 1.355 explaining 4.517% of the variance, and a Cronbach's alpha of 0.851. The fifth factor—*Autonomy*—included five items with loadings between 0.457 and 0.794, a notably high eigenvalue of 11.458, accounting for 38.194% of the variance, and a Cronbach's alpha of 0.829. Although this eigenvalue is substantially larger than those of the other factors, it reflects the central role of autonomy as a foundational competence underlying teachers' engagement with AI in educational contexts. In the context of preservice teacher education, autonomy-related items, such as independent decision-making, self-directed learning of AI tools, and confidence in managing AI use, tend to show strong intercorrelations, which can result in a dominant first-order factor during exploratory analysis. Importantly, this interpretation is supported by the theoretical positioning of autonomy within the ED-AI framework, where agency and self-regulation are conceptualized as enabling conditions for other AI-related competencies. The sixth factor—*Ethics*—also comprised five items, with loadings ranging from 0.596 to 0.794, an eigenvalue of 2.224 explaining 7.412% of the variance, and a Cronbach's alpha of 0.867. Overall, the six-factor model explained 63.965% of the total variance, with each factor demonstrating acceptable to strong internal consistency, as indicated by Cronbach's alpha values ranging from 0.759 to 0.867.

4.3. Confirmatory factor analysis

After identifying the factor structure of the TAILS through EFA, we conducted a CFA using the remaining subsample to validate the model. Model fit was assessed utilizing multiple indices, encompassing Chi-square/df, RMSEA, SRMR, TLI, and CFI (see Table 5). The initial CFA,

Table 5
Model of fit of the TAILS based on CFA results (N = 227).

Fit Indices	Recommended value	CFA models	Model values	Goodness of fit
ChiSqr/df	<5.0	CFA model 1 (based on EFA)	1.881	Good fit
		Refined CFA model 2	1.918	Good fit
		Refined CFA model 3	1.766	Good fit
RMSEA	<0.08	CFA model 1 (based on EFA)	0.062	Good fit
		Refined CFA model 2	0.064	Good fit
		Refined CFA model 3	0.058	Good fit
SRMR	<0.08	CFA model 1 (based on EFA)	0.051	Good fit
		Refined CFA model 2	0.051	Good fit
		Refined CFA model 3	0.054	Good fit
TLI	>0.90	CFA model 1 (based on EFA)	0.881	Unsatisfactory fit
		Refined CFA model 2	0.885	Unsatisfactory fit
		Refined CFA model 3	0.908	Good fit
CFI	>0.90	CFA model 1 (based on EFA)	0.893	Unsatisfactory fit
		Refined CFA model 2	0.898	Unsatisfactory fit
		Refined CFA model 3	0.919	Good fit

based on the EFA-derived 30-item model, demonstrated a good fit according to the Chi-square/df, RMSEA, and SRMR values (Kline, 2016). Nonetheless, the CFI (0.893) and TLI (0.881) approached, but did not fully reach, the recommended 0.90 threshold, indicating potential areas for model refinement.

Model modification decisions were guided by both statistical evidence and theoretical considerations rather than fit improvement alone. Examination of standardized factor loadings revealed that two items (CON2 and EVA5) loaded weakly (<0.60) on their intended constructs, suggesting limited contribution to construct representation. Conceptually, item CON2 emphasized general support for students with special needs, which overlapped with the knowledge and autonomy dimensions rather than uniquely capturing contextual adaptation, while item EVA5 focused on fairness in AI-based assessment, a concept more centrally aligned with the ethics dimension. Their removal therefore reduced construct ambiguity and improved conceptual distinctiveness without compromising core content coverage. The revised model yielded a modest improvement in fit (CFI = 0.898; TLI = 0.885). Further inspection identified one additional item (COL2) with a loading below 0.60, which was removed in the subsequent model. Item COL2 emphasized general teamwork support rather than explicit human-AI collaboration, making it less theoretically central to the collaboration construct as defined in the ED-AI framework. Its removal strengthened the construct's focus on pedagogically meaningful AI-mediated collaboration. Importantly, each affected dimension retained at least four conceptually coherent items, ensuring adequate coverage of the underlying construct. Modification indices further indicated substantial shared residual variance between two ethics items (ETH1 and ETH2). Rather than relying solely on statistical criteria, the decision to correlate these error terms was guided by clear theoretical and substantive considerations grounded in the content of the items. ETH1 and ETH2 both reflect closely related aspects of ethical responsibility in AI-mediated teaching. Specifically, decisions about selecting AI tools are inherently linked to considerations of data privacy, as teachers must evaluate how AI systems collect, store, and process student data before adopting them for classroom use. From a pedagogical and ethical standpoint, these judgments are often made concurrently within teachers' ethical decision-making processes, making some degree of shared unexplained variance conceptually plausible. Importantly, this covariance does not indicate item redundancy or construct misspecification: ETH1 focuses on the enactment of ethical awareness during AI use, whereas ETH2 captures anticipatory ethical judgment during AI selection. Allowing the residuals to correlate therefore acknowledges shared contextual specificity and overlapping ethical considerations while preserving the conceptual integrity and unidimensionality of the Ethics construct. This approach is consistent with recommended practices in structural equation modelling when residual correlations are theoretically justified and transparently reported (Brown, 2015).

The third and final CFA model yielded a refined 27-item version of the TAILS (see Appendix 1), demonstrating enhanced and satisfactory model fit (CFI = 0.919; TLI = 0.908). The final scale encompasses five items each for the *Knowledge*, *Autonomy*, and *Ethics* domains and four items each for the *Evaluation*, *Collaboration*, and *Contextualization* domains (see Fig. 2). One residual covariance pair (ETH1–ETH2) was allowed based on modification indices (Brown, 2015).

Additionally, Fig. 2 illustrates the standardized results of the six-factor model, presenting the factor loadings for the six latent constructs: *Knowledge*, *Evaluation*, *Collaboration*, *Contextualization*, *Autonomy*, and *Ethics*. For the *Knowledge* construct, factor loadings ranged from 0.601 to 0.793, demonstrating good associations. The *Evaluation* construct exhibited loadings between 0.619 and 0.714, reflecting robust relationships. Loadings for the *Collaboration* construct ranged from 0.665 to 0.712, proposing strong correlations. The *Contextualization* construct revealed similarly strong loadings, ranging from 0.685 to 0.781. For the *Autonomy* construct, the loadings ranged from 0.645 to 0.730, while the *Ethics* construct displayed loadings between 0.635 and

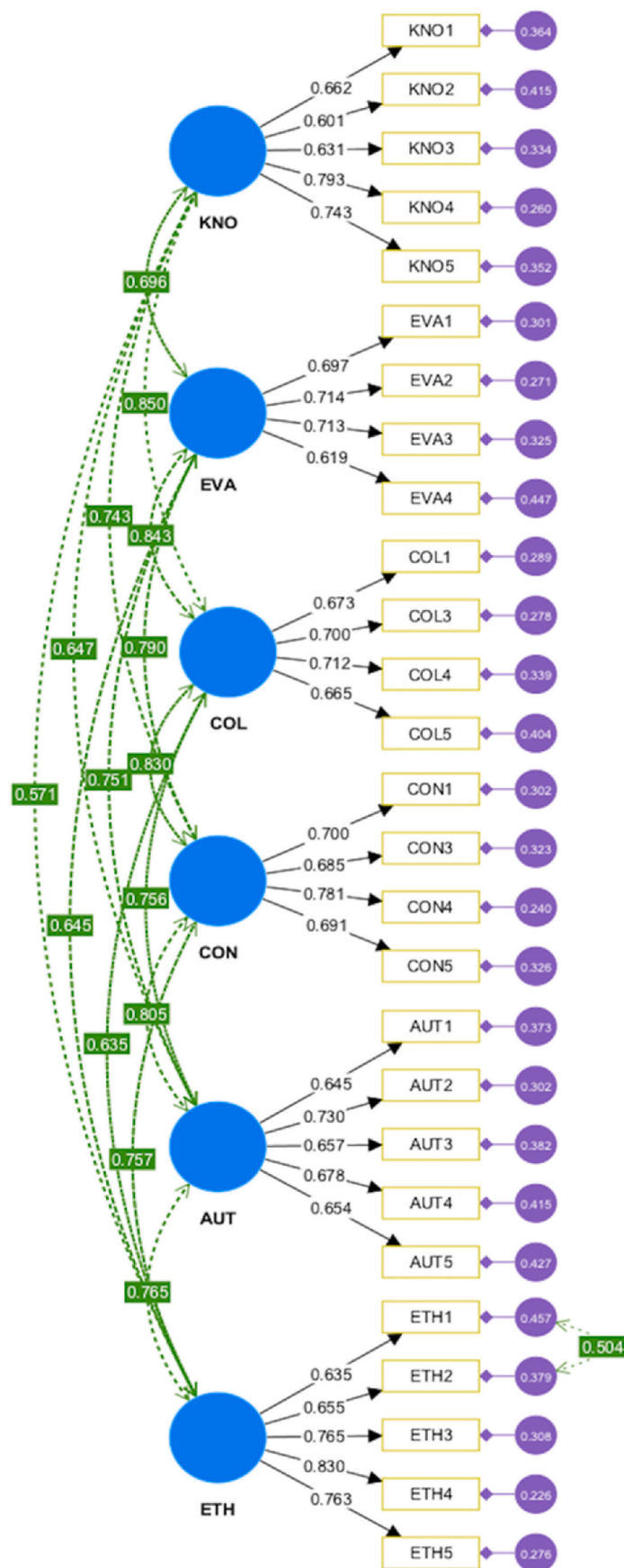


Fig. 2. Six-factor structural model of the TAILS.

Note. KNO = Knowledge; EVA = Evaluation; COL = Collaboration; CON = Contextualization; AUT = Autonomy; ETH = Ethics.

0.830, both of which were considered robust. All factor loadings were statistically significant, suggesting strong associations between the observed items and their corresponding underlying constructs.

Overall, the item deletions and residual correlation improved construct clarity and theoretical alignment rather than merely enhancing statistical fit. The refined TAILS retains comprehensive coverage of all six dimensions of the ED-AI literacy framework, ensuring that each construct is represented by a focused yet sufficient set of indicators. These results support the conceptual integrity, psychometric robustness, and practical applicability of the finalized scale.

4.4. Reliability

The internal consistency and reliability of the TAILS were assessed utilizing Cronbach's alpha and composite reliability (CR) for each of its six dimensions. All dimensions demonstrated excellent reliability, with values well above the recommended threshold of 0.70. As shown in Table 6, the *Contextualization* dimension has the highest reliability, with a Cronbach's alpha of 0.949 and a CR of 0.935. This was followed by the *Ethics* ($\alpha = 0.945$, CR = 0.949) and *Collaboration* ($\alpha = 0.925$, CR = 0.937) dimensions. The *Evaluation* and *Knowledge* dimensions each had a Cronbach's alpha of 0.920, with CRs of 0.925 and 0.941, respectively. Finally, the *Autonomy* dimension demonstrated a Cronbach's alpha of 0.903 and a CR of 0.940. These findings demonstrate that each dimension of the scale possesses robust internal consistency and composite reliability, reinforcing the robustness of the TAILS as a tool for assessing AI literacy in teacher education.

4.5. Validity

To confirm the scale's convergent validity, AVE values were calculated to examine the extent to which items within each construct are correlated. According to Fornell and Larcker (1981), convergent validity should be examined using three criteria: (1) internal consistency reliability (Cronbach's alpha), (2) CR, and (3) AVE. As presented in Table 6, the AVE values for the six dimensions of the TAILS ranged from 0.454 to 0.538, calculated from the standardized factor loadings of the finalized 27-item measurement model. Specifically, *Ethics* (0.538) and *Contextualization* (0.512) exceeded the commonly recommended threshold of 0.50, indicating that these constructs explain more than half of the variance in their respective indicators. The remaining dimensions—*Knowledge* (0.476), *Evaluation* (0.472), *Collaboration* (0.473), and *Autonomy* (0.454)—yielded AVE values slightly below the 0.50 benchmark. However, all constructs demonstrated strong CR (all exceeding 0.70, as proposed by Nunnally & Bernstein, 1994) and high Cronbach's alpha values (all above 0.70, as recommended by Hair et al., 2006), suggesting satisfactory internal consistency. Consistent with methodological guidance of Fornell and Larcker (1981), when CR is high, AVE values marginally below 0.50 may still indicate adequate convergent validity. Taken together, these results provide acceptable and theoretically coherent evidence of convergent validity for the TAILS measurement model.

Discriminant validity, which assesses whether items within a construct are more strongly associated with each other than with items from other constructs, was explored utilizing the Heterotrait–Monotrait

ratio of correlations (HTMT), as recommended by Henseler et al. (2015). The results presented in Table 7 confirm that all HTMT values ranged from 0.613 to 0.874, well below the accepted threshold of 0.90 (Gold et al., 2001), thereby indicating satisfactory discriminant validity. The six dimensions represent related facets of teacher AI literacy, which support interpreting HTMT values below 0.90 as acceptable separation in a multidimensional scale (Lim et al., 2026). One HTMT value approached this benchmark, which signals close proximity between the relevant dimensions and motivates future item refinement and replication checks in new samples. To further strengthen the evidence for discriminant validity, bootstrapped 95% confidence intervals for the HTMT estimates were examined using 1000 resamples. As shown in Table 7, none of the upper bounds of the HTMT confidence intervals exceeded the critical value of 1.00, indicating that all latent constructs are empirically distinct. This result provides additional robustness beyond point estimates alone and reduces concerns regarding potential construct overlap among the six dimensions of the TAILS.

Additionally, Pearson correlations were calculated between the six subscales of the finalized 27-item TAILS ($N = 227$) and two established measures: AI self-efficacy (Ayanwale et al., 2024; Carolus et al., 2023) and attitude toward AI questionnaires (Losada et al., 2023; Nordlöf, Hallström, & Höst, 2019; Yue et al., 2024) to assess concurrent criterion validity (see Table 8). While these three scales investigate related but distinct constructs, the TAILS was anticipated to correlate positively with both AI self-efficacy and attitude toward AI. As exhibited in Table 8, all correlations were statistically significant ($p < 0.01$). To aid interpretation, correlation magnitudes were evaluated using Cohen's (1988) effect size guidelines, where coefficients of approximately 0.10 indicate small, 0.30 moderate, and 0.50 or above large associations. The intercorrelations among the six TAILS subscales ranged from 0.530 to 0.695, corresponding to large effect sizes, indicating strong and theoretically coherent relationships among the dimensions while still reflecting their conceptual distinctiveness. Correlations between each TAILS dimension and AI self-efficacy ranged from 0.476 to 0.661, falling within the moderate-to-large range, while correlations with attitudes toward AI ranged from 0.491 to 0.690, also reflecting moderate-to-large effect sizes. These results offer evidence for the good concurrent criterion validity of the TAILS, exhibiting meaningful associations with established measures that assess related yet distinct constructs.

5. Discussion

This research sought to develop and validate the TAILS, grounded in the ED-AI literacy framework (Allen & Kendeu, 2024) and to assess its psychometric properties within the context of English language teacher education. The results provide strong evidence that the TAILS is a reliable and valid instrument for exploring preservice English language teachers' AI literacy across six key dimensions: *Knowledge*, *Evaluation*, *Collaboration*, *Contextualization*, *Autonomy*, and *Ethics*. The psychometric evaluation—including both exploratory and confirmatory factor analyses—revealed a six-factor structure that aligned well with the theoretical framework. All dimensions exhibited high internal consistency (Cronbach's $\alpha > 0.90$) and appropriate model fit indices in the final CFA (CFI = 0.919, TLI = 0.908, RMSEA = 0.058), meeting the psychometric standards recommended by Kline (2016) and Hu and Bentler (1999).

Table 6

Internal consistency reliability (CRB), composite reliability (CR), and average variance extracted (AVE).

Dimensions	Cronbach's alpha	Composite reliability (rho_c)	Average variance extracted (AVE)
1. Knowledge	0.920	0.941	0.476
2. Evaluation	0.920	0.925	0.472
3. Collaboration	0.925	0.937	0.473
4. Contextualization	0.949	0.935	0.512
5. Autonomy	0.903	0.940	0.454
6. Ethics	0.945	0.949	0.538

Table 7
HTMT ratio and confidence interval.

Dimensions	1	2	3	4	5	6
1.Knowledge						
2.Evaluation	0.725 [0.600-0.817]					
3.Collaboration	0.874 [0.770-0.954]	0.855 [0.748-0.933]				
4.Contextualization	0.756 [0.629-0.854]	0.810 [0.706-0.898]	0.848 [0.733- 0.929]			
5.Autonomy	0.661 [0.503-0.771]	0.753 [0.626- 0.857]	0.762 [0.651- 0.851]	0.820 [0.708-0.904]		
6.Ethics	0.633 [0.507-0.762]	0.649 [0.518- 0.773]	0.668 [0.528- 0.787]	0.790 [0.666-0.893]	0.783 [0.663- 0.875]	

Note. The confidence interval of the value is in parentheses.

Table 8
Means, standard deviations, and correlations among TAILS scores, AI self-efficacy questionnaire scores, and attitude toward AI questionnaire (N = 227).

Dimensions	M	SD	1	2	3	4	5	6	7	8
1.Knowledge	3.989	0.620								
2.Evaluation	3.667	0.616	0.576 ^a							
3.Collaboration	3.775	0.613	0.695 ^a	0.665 ^a						
4.Contextualization	3.754	0.620	0.611 ^a	0.640 ^a	0.669 ^a					
5.Autonomy	3.697	0.626	0.535 ^a	0.595 ^a	0.603 ^a	0.658 ^a				
6.Ethics	3.740	0.677	0.530 ^a	0.530 ^a	0.547 ^a	0.656 ^a	0.651 ^a			
7.AI Self-efficacy	3.667	0.640	0.478 ^a	0.476 ^a	0.565 ^a	0.661 ^a	0.566 ^a	0.518 ^a		
8.Attitude toward AI	3.946	0.661	0.602 ^a	0.491 ^a	0.690 ^a	0.623 ^a	0.558 ^a	0.545 ^a	0.540 ^a	

^a Correlation is significant at the 0.01 level (2-tailed).

Although Cronbach's alpha values exceeding 0.90 may, in some cases, suggest potential item redundancy, such values are considered acceptable in the context of newly developed, theory-driven instruments designed to measure complex and multidimensional constructs (Boateng et al., 2018; DeVellis & Thorpe, 2021). In the present study, the high alpha coefficients reflect the conceptual coherence and close alignment of items within each dimension of the ED-AI literacy framework rather than unnecessary duplication. This interpretation is further supported by the satisfactory CR and AVE values across all dimensions, as well as by evidence of discriminant validity, indicating that the constructs are empirically distinct. Moreover, the iterative scale development process, including expert review, pilot testing, EFA-based refinement, and CFA-driven item reduction, helped ensure that retained items contributed uniquely and meaningfully to their respective constructs. Therefore, the observed high internal consistency can be understood as an indicator of measurement precision and construct clarity.

The successful operationalization of the ED-AI framework into a psychometrically sound tool addresses a crucial gap in the field. Previous AI literacy instruments, such as the AILQ, AILS, and MAILS, have primarily targeted students (Ng et al., 2023), general users (Wang et al., 2022), or non-discipline-specific audiences (Carolus et al., 2023), offering limited applicability to English language teacher education. Notably, the TAILS instrument has a significant strength owing to its contextual specificity. While earlier studies (e.g., Chu et al., 2023; Shah, 2023) have highlighted the growing need for AI integration in language classrooms, this research is the first to offer an empirically validated tool tailored specifically for language educators to assess their readiness and capacity for effective AI integration.

Each dimension of the TAILS scale reflects the essential competencies that are aligned with contemporary demands in AI-integrated English language teaching. The *Knowledge* dimension, which examines teachers' foundational understanding of AI concepts and their applications in education, corresponds with previous research proposing that preservice teachers are increasingly aware of the potential of AI tools, such as automated scoring systems and chatbots, to optimize teaching and learning (Fu et al., 2020; Kim et al., 2021). Nonetheless, as Chiu et al. (2024) argued, understanding AI must go beyond familiarity with its tools to encompass knowledge of the underlying principles and mechanisms that govern AI systems. Furthermore, the *Evaluation* dimension, which tests the ability to critically assess and select appropriate AI tools for pedagogical purposes, is consistent with the concerns raised by Lee

and Perret (2022), who noted that many teacher training programs lack structured frameworks for evaluating the effectiveness, fairness, and curricular relevance of AI technologies. Teachers must not only be proficient in using AI tools but also in scrutinizing their decision-making processes and educational implications. As Ng et al. (2023) asserted, such critical literacy is crucial to preventing the uncritical adoption of AI and ensuring its meaningful integration into instruction.

Additionally, the *Collaboration* dimension, which reflects the extent to which teachers can leverage AI to optimize learner collaboration, reinforces the work of Fan and Zhang (2024), who reported that AI-supported collaborative learning can increase student engagement and enjoyment. This is specifically relevant in language education, where interactive and communicative learning environments are central to skill development. Nonetheless, as Chu et al. (2023) cautioned, AI should serve to enrich rather than replace human interaction, ensuring that authentic communication remains at the core of language learning. The inclusion of the *Contextualization* and *Autonomy* dimensions, which are often overlooked in existing tools, illustrates the nuanced demands placed on language teachers. Educators are anticipated not only to understand and employ AI but also to adapt them to diverse linguistic, cultural, and pedagogical contexts (AlAfnan, 2024; Gyawali & Mehandroo, 2022). The *Contextualization* dimension, which measures the ability to tailor AI tools to diverse learner needs and educational environments, aligns with the warnings of AlAfnan (2024) and Gyawali and Mehandroo (2022), who noted that a failure to contextualize technology can lead to inequitable or ineffective outcomes. Meanwhile, the *Autonomy* dimension evaluates teachers' confidence in independently managing and integrating AI tools in their teaching. Unlike other instruments such as the AILQ (Ng et al., 2023) and MAILS (Carolus et al., 2023), which underscore affective or cognitive aspects, the TAILS offers a pedagogically grounded perspective with direct implications for instructional design. As Otero et al. (2023) and Ibrahim (2024) highlighted, autonomy fosters both self-efficacy and sustainable professional growth in digitally mediated teaching contexts. Finally, the *Ethics* dimension, which explores teachers' sensitivity to ethical concerns correlated with AI use, emerged as a particularly significant factor in the current digital landscape. This finding is consistent with the views of Choi et al. (2024) and Biagini (2024), who highlighted the need to prepare educators to navigate fairness, accountability, bias, and student privacy in AI-mediated environments. Nevertheless, following Chiu et al. (2024), this research affirmed that ethical competencies must not

be assumed to emerge organically from AI use. Instead, they must be explicitly and systematically scaffolded within teacher education programs.

Overall, this work makes several meaningful contributions to the field. It introduces a validated, theory-informed tool for assessing AI literacy specifically in English language teacher education—an area largely underserved by existing generalized tools. Additionally, the research empirically supports the multidimensional nature of AI literacy, emphasizing the significance of ethical awareness, contextual adaptability, and collaborative competence for effective AI integration. Moreover, the TAILS instrument holds practical value: it can inform teacher training curricula, guide professional development initiatives, and support educational policy aimed at preparing educators for AI-enhanced classrooms.

5.1. Strengths, limitations, and future research directions

One of the key strengths of this study lies in its rigorous methodological approach, particularly the integration of exploratory and confirmatory factor analyses, which ensured the robustness and structural validity of TAILS. Grounded in the ED-AI literacy framework (Allen & Kendeou, 2024), the instrument addresses a notable gap in the literature by providing a psychometrically validated, theory-driven tool specifically tailored to English language teacher education. From an educational perspective, the study advances innovation in language teacher education by operationalizing the ED-AI literacy framework into a practical assessment instrument that reflects the professional and pedagogical realities of language teachers, rather than focusing on students or general users as in most existing AI literacy measures. This shift represents a move from generic technological competence toward pedagogically grounded AI literacy. Moreover, the multidimensional structure of the TAILS enables data-informed curriculum development and personalized teacher preparation by identifying preservice teachers' strengths and needs across key dimensions. As such, teacher education programs can design targeted instructional interventions that foster more reflective, responsible, and context-sensitive AI integration, positioning the study as a contribution not only to measurement development but also to evidence-based and adaptive innovation in AI-enhanced teaching and learning environments.

Despite these strengths, several limitations should be acknowledged. First, data were collected using self-report measures, which may be subject to social desirability bias and respondents' self-perceptions of their AI literacy rather than observed instructional practices. Second, a marker-variable test was not conducted to quantify potential common method variance, which leaves residual uncertainty regarding method-driven covariance in the observed relationships. Third, the study focused exclusively on preservice teachers, which limits the extent to which the findings can be generalized to in-service or experienced language teachers whose professional responsibilities and exposure to AI tools may differ substantially. Fourth, the sample was drawn from an Indonesian context, and although this provides valuable insight into AI literacy development, the cultural, institutional, and technological conditions of teacher education in Indonesia may differ from those in other regions. As a result, caution is warranted when generalizing the findings to different national or cross-cultural contexts. Fifth, the pilot study involved a relatively small sample, which, although sufficient for establishing face validity and item clarity, limits the generalizability of the pilot findings. The pilot sample size was determined based on saturation, as no substantial new issues regarding item comprehension or clarity emerged; however, given the cross-cultural and context-sensitive nature of AI literacy, a larger and more diverse pilot sample could have provided additional insights. Sixth, the cross-sectional design restricts the ability to examine how teachers' AI literacy develops over time. Finally, Despite the comprehensive validity evidence presented, some aspects of validity remain untested. Specifically, the study did not examine nomological validity in terms of how TAILS relates to broader

teacher competency frameworks, nor did it assess predictive validity.

Future research should validate TAILS with in-service language teachers and employ longitudinal designs to examine the stability and developmental sensitivity of AI literacy across career stages. Cross-cultural measurement invariance testing across different countries and educational systems is also necessary to support meaningful international comparisons. Future research should test measurement invariance across key subgroups such as gender, academic level, and teaching experience before comparing scores across groups. Evidence for configural, metric, and scalar invariance would support meaningful subgroup comparisons. Future research should also evaluate whether the six dimensions can be modeled as a higher-order teacher artificial intelligence literacy construct using hierarchical component modeling, which would clarify whether reporting both dimension scores and an overall score is justified. Incorporating multi-method approaches beyond self-report would further strengthen the instrument's validity and explanatory power. Finally, future studies are recommended to examine the nomological validity of TAILS by situating it within broader teacher competency frameworks and to establish its predictive validity by investigating its ability to predict teachers' instructional practices and professional outcomes related to AI use.

5.2. Implications

The findings of this study have important implications for teacher education, curriculum development, educational policy, and pedagogical theory. By providing a validated, context-specific instrument for assessing AI literacy in language teacher education, the TAILS offers teacher educators a practical tool to diagnose preservice teachers' strengths and areas for development across multiple dimensions. Within teacher education programs, the TAILS can inform curriculum design by aligning course objectives, learning activities, and assessments with AI literacy dimensions such as evaluation, contextualization, collaboration, autonomy, and ethics, thereby supporting the systematic integration of pedagogically grounded AI competencies. The scale may be used diagnostically at program entry and evaluatively following AI-focused coursework to monitor learning gains, guide curriculum refinement, and support targeted professional development. At the program and policy levels, aggregated TAILS data enable data-driven decision-making for embedding AI literacy within teacher education frameworks and for evaluating trends across cohorts, institutions, or demographic groups. Beyond assessment, the multidimensional data generated by TAILS support educational and pedagogical data analytics, including longitudinal tracking of teachers' AI literacy development over time. Theoretically, the scale aligns with constructivist and sociocultural learning perspectives by emphasizing critical evaluation, contextualized AI use, collaboration, teacher agency, and ethical responsibility. Collectively, these implications position AI literacy as a pedagogically grounded, reflective, and values-oriented construct, supporting the preparation of adaptive and responsible language educators in AI-enhanced learning environments.

6. Conclusion

This research successfully developed and validated TAILS, a comprehensive and reliable instrument grounded in the ED-AI literacy framework, specifically designed for language teacher education. The scale's six-factor structure, which comprises *Knowledge*, *Evaluation*, *Collaboration*, *Contextualization*, *Autonomy*, and *Ethics*, reflects the multifaceted competencies essential for navigating AI-enhanced teaching and learning environments. The robust psychometric evidence affirms its potential for use in both research and practice, offering a valuable tool for assessing preservice teachers' AI literacy and informing teacher training programs. This study addresses a critical gap in existing AI literacy assessments and contributes meaningfully to the advancement of teacher preparedness in the age of AI. Future research should

explore the applicability of the TAILS in broader educational and cultural contexts.

CRedit authorship contribution statement

Salim Nabhan: Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anita Habók:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Ethical approval

The project was approved by IRB at the Doctoral School of Education, University of Szeged, Hungary, with the reference number January 2025. Written consent to participate was obtained from all participants prior to the study, and their privacy rights were strictly observed.

Declaration on the use of AI statement

No content generated by AI technologies has been used in the project work.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix 1. The table below exhibits the originally administered 30-item scale. Based on the findings of the CFA, three items (CON2, EVA5, and COL2) were removed, and a pair of items (ETH1 and ETH2) was allowed to correlate. Thus, the final model comprised a refined 27-item version of the TAILS

Teachers' AI Literacy Scale (TAILS)		
Codes	Items	Removal & Correlation based on CFA
Knowledge (KNO)		
KNO1	1. I know that AI can help teachers create personalized learning for students.	
KNO2	2. I am familiar with some common AI tools that support teaching and learning in classrooms (e.g., ChatGPT).	
KNO3	3. I know that AI can support students with different learning needs.	
KNO4	4. I know that AI can assist teachers in lesson planning and material development.	
KNO5	5. I am aware that AI applications can help teachers with assessment.	
Evaluation (EVA)		
EVA1	6. I am capable of comparing the effectiveness of different AI tools for classroom use.	
EVA2	7. I can select the most appropriate solution from different options provided by AI applications.	
EVA3	8. I can choose the right educational AI tools based on the specific task.	
EVA4	9. I can evaluate how AI recommendations align with curriculum standards and goals.	
EVA5	10. I can analyze whether AI assessment tools for students are accurate and fair.	Remove
Collaboration (COL)		
COL1	11. I can collaborate with AI tools to design engaging lessons.	
COL2	12. I can use AI to support teamwork among students in group projects.	Remove
COL3	13. I can communicate effectively with AI tools to modify group learning activities based on student needs.	
COL4	14. I feel comfortable integrating feedback from AI systems into collaborative lesson planning.	
COL5	15. I can facilitate student collaboration using AI-enhanced platforms (e.g., Padlet).	
Contextualization (CON)		
CON1	16. I can integrate AI tools into lesson plans to enhance specific subjects, such as language learning.	
CON2	17. I understand how to use AI to support students with specific needs.	Remove
CON3	18. I can design classroom activities that encourage students to explore AI technologies.	
CON4	19. I can apply AI tools to improve teaching strategies.	
CON5	20. I can adapt AI applications to suit cultural contexts in my teaching.	
Autonomy (AUT)		
AUT1	21. I can decide when it is appropriate to use AI tools in my teaching and when not to.	
AUT2	22. I am capable of learning new AI tools independently for teaching purposes.	
AUT3	23. I can troubleshoot basic technical issues with AI tools used in the classroom.	
AUT4	24. I am confident in guiding students on how to interact with AI responsibly.	
AUT5	25. I feel prepared to design lessons without relying solely on AI, when necessary.	
Ethics (ETH)		
ETH1	26. I actively consider students' privacy when using AI tools.	Correlate
ETH2	27. I prioritize ethical considerations when selecting AI tools for classroom use.	Correlate
ETH3	28. I can quickly identify ethical issues when using AI applications for my teaching.	
ETH4	29. I can identify and address potential fairness issues when using AI for assessment.	
ETH5	30. I feel responsible for ensuring that AI use in my teaching activities is ethical.	

Appendix 2. Content Validity Index and Modified Kappa

Items	Expert 1	Expert 2	Expert 3	Expert 4	Expert 5	Expert 6	Expert 7	Total no. of experts (N)	Total no. of experts in agreement (A)	N-A	UA	I-CVI	P _c	Modified Kappa (κ*)	Interpretation
KNO1	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
KNO2	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
KNO3	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
KNO4	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
KNO5	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
EVA1	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
EVA2	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
EVA3	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
EVA4	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
EVA5	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
COL1	1	1	0	1	1	1	1	7	6	1	0	0.9	0.0068359	0.86	Excellent
COL2	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
COL3	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
COL4	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
COL5	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
CON1	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
CON2	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
CON3	1	1	0	1	1	1	1	7	6	1	0	0.9	0.0068359	0.86	Excellent
CON4	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
CON5	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
AUT1	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
AUT2	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
AUT3	1	1	1	1	1	0	1	7	6	1	0	0.9	0.0068359	0.86	Excellent
AUT4	1	1	1	1	1	0	1	7	6	1	0	0.9	0.0068359	0.86	Excellent
AUT5	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
ETH1	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
ETH2	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
ETH3	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
ETH4	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
ETH5	1	1	1	1	1	1	1	7	7	0	1	1	0.0009766	1	Excellent
											S-CVI	29.4			
											S-CVI/Ave	0.98			
											SCVI/UA	0.87			

Note. KNO = knowledge; EVA = evaluation; COL = collaboration; CON = contextualization; AUT = autonomy; ETH = ethics; CVI = content validity index; S-CVI = scale-level CVI; S-CVI/Ave = scale-level CVI/average; S-CVI/UA = scale-level CVI/universal agreement; P_c = probability of chance agreement.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

AlAfnan, M. A. (2024). Artificial intelligence and language: Bridging Arabic and English with technology. *Journal of Ecohumanism*, 3(8), 240–256. <https://doi.org/10.62754/joe.v3i8.4961>

Allen, L. K., & Kendeou, P. (2024). ED-AI lit: An interdisciplinary framework for AI literacy in education. *Policy Insights from the Behavioral and Brain Sciences*, 11(1), 3–10. <https://doi.org/10.1177/23727322231220339>

Ayanwale, M. A., Adelana, O. P., Molefi, R. R., Adeeko, O., & Ishola, A. M. (2024). Examining artificial intelligence literacy among pre-service teachers for future classrooms. *Computers and Education Open*, 6, Article 100179. <https://doi.org/10.1016/j.caeo.2024.100179>

Biagini, G. (2024). Assessing the assessments: Toward a multidimensional approach to AI literacy. *Media Education*, 15(1), 91–101. <https://doi.org/10.36253/me-15831>

Boateng, G. O., Neilands, T. B., Frongillo, E. A., Melgar-Quinonez, H. R., & Young, S. L. (2018). Best practices for developing and validating scales for health, social, and behavioral research: A primer. *Frontiers in Public Health*, 6, 149. <https://doi.org/10.3389/fpubh.2018.00149>

Boscardin, C., Gin, B., Golde, P., & Hauer, K. (2023). Chatgpt and generative artificial intelligence for medical education: Potential impact and opportunity. *Academic Medicine*, 99(1), 22–27. <https://doi.org/10.1097/acm.0000000000005439>

Brown, T. A. (2015). *Confirmatory Factor Analysis for Applied Research* (Second edition). The Guilford Press.

Carolus, A., Koch, M. J., Straka, S., Latoschik, M. C., & Wienrich, C. (2023). MAILS – Meta AI literacy scale: Development and testing of an AI literacy questionnaire based on well-founded competency models and psychological change- and meta-competencies. *Computers in Human Behavior: Artificial Humans*, 1(2), 1–10. <https://doi.org/10.1016/j.chbah.2023.100014>

Chen, Q., Tawilapakul, U., & Lin, A. M. Y. (2026). Teacherness in the age of GenAI: An ethnographic exploration of pedagogical decision-making and AI integration in English language teaching. *System*, 136, Article 103877. <https://doi.org/10.1016/j.system.2025.103877>

Chiu, T. K. F., Ahmad, Z., Ismailov, M., & Sanusi, I. T. (2024). What are artificial intelligence literacy and competency? A comprehensive framework to support them. *Computers and Education Open*, 6, Article 100171. <https://doi.org/10.1016/j.caeo.2024.100171>

Choi, J., Yang, E., & Goo, E. (2024). The effects of an ethics education program on artificial intelligence among middle school students: Analysis of perception and attitude changes. *Applied Sciences*, 14(4), 1588. <https://doi.org/10.3390/app14041588>

Chu, T., Wang, W., & Chen, X. (2023). Visual analysis of research trends on language teachers in the era of artificial intelligence. *Advances in Educational Technology and Psychology*, 7(8). <https://doi.org/10.23977/aetp.2023.070812>

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Hillsdale, NJ: Lawrence Erlbaum Associates, Publishers.

DeVellis, R. F., & Thorpe, C. T. (2021). *Scale development: Theory and applications*. Thousand Oaks: Sage Publications.

Fan, J., & Zhang, Q. (2024). From literacy to learning: The sequential mediation of attitudes and enjoyment in AI-assisted EFL education. *Heliyon*, 10(17), Article e37158. <https://doi.org/10.1016/j.heliyon.2024.e37158>

Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18, 39–50. <https://doi.org/10.2307/3151312>

Fu, S., Gu, H., & Yang, B. (2020). The affordances of AI-enabled automatic scoring applications on learners' continuous learning intention: An empirical study in China. *British Journal of Educational Technology*, 51(5), 1674–1692. <https://doi.org/10.1111/bjet.12995>

Gold, A. H., Malhotra, A., & Segars, A. H. (2001). Knowledge management: An organizational capabilities perspective. *Journal of Management Information Systems*, 18(1), 185–214. <https://doi.org/10.1080/07421222.2001.11045669>

Gyawali, Y., & Mehandroo, M. (2022). Artificial intelligence in English language teaching: Navigating the future with emerging perspectives. *Journal of Language and Linguistics in Society*, 2(6), 21–27. <https://doi.org/10.55529/jlls.26.21.27>

- Hair, J. F., Black, W. C., Babin, B. J., Anderson, R. E., & Tatham, R. L. (2006). *Multivariate data analysis* (6th ed.). Upper Saddle River, NJ: Pearson University Press.
- Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, 43, 115–135. <https://doi.org/10.1007/s11747-014-0403-8>
- Hollebeck, L. D., Kumar, V., Srivastava, R. K., Lim, W. M., & Urbonavicius, S. (2025). Guidelines for theory selection: The IMPACT framework. *Psychology and Marketing*, 42(11), 2789–2806. <https://doi.org/10.1002/mar.70009>
- Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal*, 6(1), 1–55. <https://doi.org/10.1080/10705519909540118>
- Ibrahim, A. (2024). Assessing the knowledge and perception of artificial intelligence for teaching and research among lecturers in the faculties of arts in Nigeria. *Journal of Global Research in Education and Social Science*, 18(2), 25–33. <https://doi.org/10.56557/jogress/2024/v18i28671>
- Joseph, S. (2023). Large language model-based tools in language teaching to develop critical thinking and sustainable cognitive structures. *Rupkatha Journal on Interdisciplinary Studies in Humanities*, 15(4). <https://doi.org/10.21659/rupkatha.v15n4.13>
- Kim, H., Cha, Y., & Kim, N. Y. (2021). Effects of AI chatbots on EFL students' communication skills. *Korean Journal of English Language and Linguistics*, 21, 712–734. <https://doi.org/10.15738/kjell.21..202108.712>
- Kline, R. B. (2016). *Principles and practice of structural equation modeling* (4th ed.). The Guilford Press.
- Laupichler, M. C., Aster, A., & Raupach, T. (2023). Delphi study for the development and preliminary validation of an item set for the assessment of non-experts' AI literacy. *Computers and Education: Artificial Intelligence*, 4, Article 100126. <https://doi.org/10.1016/j.caeai.2023.100126>
- Lee, I., & Perret, B. (2022). Preparing high school teachers to integrate AI methods into STEM classrooms. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(11), 12783–12791. <https://doi.org/10.1609/aaai.v36i11.21557>
- Lim, W. M. (2024). A typology of validity: Content, face, convergent, discriminant, nomological and predictive validity. *Journal of Trade Science*, 12(3), 155–179. <https://doi.org/10.1108/JTS-03-2024-0016>
- Lim, W. M. (2026). Theory and theory development: Guidelines for establishing theoretical gaps, foundations, contributions, and implications. *Journal of Business Research*, 202. <https://doi.org/10.1016/j.jbusres.2025.115745>
- Lim, W. M., Khatri, P., Shiva, A., & Dabas, V. (2026). Guidelines for scale development and validation. *Journal of Consumer Behaviour*, 1–21. <https://doi.org/10.1002/cb.70096>
- Liu, F., Makady, H., Nah, S., & McNealy, J. (2023). When citizens support AI policies: The moderating roles of AI efficacy on AI news, discussion, and literacy. *Journal of Information Technology & Politics*, 1–17. <https://doi.org/10.1080/19331681.2023.2294363>
- Long, D., & Magerko, B. (2020). What is AI literacy? Competencies and design considerations. In *Proceedings of the 2020 CHI conference on human factors in computing systems* (pp. 1–16). <https://doi.org/10.1145/3313831.3376727>
- Losada, D., Campo, L., Maza, M. S. D. L., & Galindo Domínguez, H. (2023). Design and validation of a multidimensional scale for assessing teachers' perceptions towards artificial intelligence in education. *International Journal of Learning Technology*, 1(1), Article 10062094. <https://doi.org/10.1504/IJLT.2023.10062094>
- Mont, M., Dooley, M., & Masats, D. (2026). Enhancing foreign language learning through AI-driven telecollaboration. *Journal of Responsible Technology*, 25, Article 100153. <https://doi.org/10.1016/j.jrt.2026.100153>
- Ng, D. T. K., Leung, J. K. L., Chu, S. K. W., & Qiao, M. S. (2021a). Conceptualizing AI literacy: An exploratory review. *Computers and Education: Artificial Intelligence*, 2, Article 100041. <https://doi.org/10.1016/j.caeai.2021.100041>
- Ng, D., Leung, J., Chu, K., & Qiao, S. (2021b). AI literacy: Definition, teaching, evaluation and ethical issues. *Proceedings of the Association for Information Science and Technology*, 58(1), 504–509. <https://doi.org/10.1002/pra2.487>
- Ng, D. T. K., Wu, W., Leung, J. K. L., & Chu, S. K. W. (2023). Artificial Intelligence (AI) literacy questionnaire with confirmatory factor analysis. In *2023 IEEE international conference on advanced learning technologies (ICALT)* (pp. 233–235). IEEE. <https://doi.org/10.1109/ICALT58122.2023.00074>
- Nordlöf, C., Hallström, J., & Höst, G. E. (2019). Self-efficacy or context dependency?: Exploring teachers' perceptions of and attitudes towards technology education. *International Journal of Technology and Design Education*, 29(1), 123–141. <https://doi.org/10.1007/s10798-017-9431-2>
- Nunnally, J. C., & Bernstein, I. H. (1994). *Psychometric theory*. New York: McGraw-Hill, Inc.
- Otero, L., Catalá, A., Fernández-Morante, C., Taboada, M., López, B., & Barro, S. (2023). AI literacy in K-12: A systematic literature review. *International Journal of STEM Education*, 10(1). <https://doi.org/10.1186/s40594-023-00418-7>
- Polit, D. F., Beck, C. T., & Owen, S. V. (2007). Focus on research method: Is the CVI an acceptable indicator of content validity? Appraisal and recommendations. *Research in Nursing & Health*, 30, 459–467. <https://doi.org/10.1002/nur.20199>
- Relmasira, S., Lai, Y., & Donaldson, J. (2023). Fostering AI literacy in elementary science, technology, engineering, art, and mathematics (STEAM) education in the age of generative AI. *Sustainability*, 15(18), Article 13595. <https://doi.org/10.3390/su151813595>
- Schepman, A., & Rodway, P. (2020). Initial validation of the general attitudes towards Artificial Intelligence Scale. *Computers in Human Behavior Reports*, 1, Article 100014. <https://doi.org/10.1016/j.chbr.2020.100014>
- Shah, P. (2023). *AI and the future of education: Teaching in the age of artificial intelligence*. Jhon Wiley & Sons, Inc.
- Sindermann, C., Sha, P., Zhou, M., Wernicke, J., Schmitt, H. S., Li, M., Sariyska, R., Becker, B., & Montag, C. (2021). Assessing the attitude towards artificial intelligence: Introduction of a short measure in German, Chinese, and English Language. *KI – Künstliche Intelligenz*, 35(1), 109–118. <https://doi.org/10.1007/s13218-020-00689-0>
- Taherdoost, H. (2016). Validity and reliability of the research instrument; how to test the validation of a questionnaire/survey in a research. *International Journal of Academic Research in Management*, 5(3), 28–36. <https://doi.org/10.2139/ssrn.3205040>
- Torres, P. J., & Kahveci, Y. E. (2025). Effectiveness of Artificial Intelligence (AI) in language teaching. *Computers and Education: Artificial Intelligence*, 9, Article 100522. <https://doi.org/10.1016/j.caeai.2025.100522>
- Wang, X., Gao, Y., Wang, Q., & Zhang, P. (2024). Fostering engagement in AI-mediate Chinese EFL classrooms: The role of classroom climate, AI literacy, and resilience. *European Journal of Education*, 60(1). <https://doi.org/10.1111/ejed.12874>
- Wang, B., Rau, P. L. P., & Yuan, T. (2022). In measuring user competence in using artificial intelligence: Validity and reliability of artificial intelligence literacy scale. *Behaviour & Information Technology*, 42(9), 1324–1337. <https://doi.org/10.1080/0144929X.2022.2072768>
- Wang, Y. Y., & Wang, Y. S. (2022). Development and validation of an artificial intelligence anxiety scale: An initial application in predicting motivated learning behavior. *Interactive Learning Environments*, 30(4), 619–634. <https://doi.org/10.1080/10494820.2019.1674887>
- Wei, L. (2023). Artificial intelligence in language instruction: Impact on English learning achievement, L2 motivation, and self-regulated learning. *Frontiers in Psychology*, 14, Article 1261955. <https://doi.org/10.3389/fpsyg.2023.1261955>
- Yeter, I., Yang, W., & Sturgess, J. (2024). Global initiatives and challenges in integrating artificial intelligence literacy in elementary education: Mapping policies and empirical literature. *Future in Educational Research*, 2(4), 382–402. <https://doi.org/10.1002/fer.3.59>
- Yetişensoy, O., & Rapoport, A. (2023). Artificial intelligence literacy teaching in social studies education. *Journal of Pedagogical Research*, 7(3), 100–110. <https://doi.org/10.3390/JPR.202320866>
- Yue, M., Jong, M. S.-Y., & Ng, D. T. K. (2024). Understanding K–12 teachers' technological pedagogical content knowledge readiness and attitudes toward artificial intelligence education. *Education and Information Technologies*, 29(15), 19505–19536. <https://doi.org/10.1007/s10639-024-12621-2>
- Yusoff, M. S. B., Arifin, W. N., & Hadie, S. N. H. (2021). ABC of questionnaire development and validation for survey research. *Education in Medicine Journal*, 13(1), 97–108. <https://doi.org/10.21315/eimj2021.13.1.10>
- Zhao, L., Wu, X., & Luo, H. (2022). Developing AI literacy for primary and middle school teachers in China: Based on a structural equation modeling analysis. *Sustainability*, 14(21), Article 14549. <https://doi.org/10.3390/su142114549>